Machine Learning Techniques for Improving Multiclass Anomaly Detection on Conveyor Belts

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***Abstract*—Industrial conveyor belt systems are an efficient means of transport due to their adaptability and extension. Nonetheless, such systems are prone to various failures, including but not limited to: idler anomalies; belt tears; and pin misalignment which can cause significant disruptions in the production process. Preemptive maintenance and health monitoring of these conveyor belts is a common practice for avoiding these failures, but a challenging task due to the rarity of comprehensive anomaly detection datasets in the area, with current works aimed at evaluating the belt's immediate condition at fixed points. This study addresses this research gap by comparing multiple machine learning techniques, such as a proposed Hybrid Neural Network (HNN) tailored for classification of multiple anomaly classes, as well as machine learning approaches for time series based on feature extraction, Catch22, Minirocket Arsenal, and Time Series Forest. Our proposed HNN architecture performed better in classifying and distinguishing between different types of anomalies, with an accuracy of 95.55%. The obtained results suggest a promising approach for the area of predictive maintenance for industrial conveyor systems, as well as gives insights on possible improvements on the model and future research.**

***Index Terms*—Inertial Measurement Unit, conveyor belt, anomaly detection, machine learning, hybrid transformers, mining industry**

I. Introduction

Industrial conveyor belt systems are essential in transporting materials, especially in the mining industry, which requires long distance transportation of heavy ores and stones [1]. However, despite the robustness of these belts, they are still prone to damage, with common faults including mistracking, slippage, tear, deviation, and idler faults [1], [2], [3]. Damage detection and maintenance of the belt structure is crucial for the efficiency of the overall system, as well as the safety of the workers involved. Traditionally, the maintenance approach for these systems has been reactive instead of preemptive, primarily due to challenges in accurately monitoring the system's health.

Various solutions are available to address the problems associated with conveyor belts, such as mechanical devices, ultrasonic and acoustic sensors, vibration analysis, and computer vision approaches. However, mechanical devices, like switches and gates, only partially mitigate the problem, and ultrasonic and acoustic sensors can only monitor a fixed number of idlers. Moreover, optical systems are heavily influenced by detrimental environmental conditions common in mining sites, characterized by dust, rain, and poor lighting conditions, which directly affect image processing accuracy [2]. Additionally, most solutions only evaluate the belt or the idler's belt condition point-wise, which is not viable for long distance belts with multiple pins and rollers. Therefore, novel methods to inspect and monitor conveyor belts are a pertinent research topic to the mining industry.

This work aims to evaluate the health of a conveyor belt by utilizing Inertial Measurement Units (IMUs) coupled to the belt to detect misalignment events in any pins throughout the conveyor belt. We used a dataset based on [3], in which reference and anomalous time-series data were collected. The experiments were realized on a reduced-scale conveyor belt under different scenarios, including a 'normal' series for reference and idlers' structural anomalies series. Hence, we analyzed the measured data to evaluate its health.

To maximize the accuracy of the anomaly classification model, our study compared various machine learning techniques. We implemented a Random Forest (RF) model and a TSForest [4] instance for initial testing. Following the training and testing of the RF models, we implemented a succession of feature extraction models: a Catch22 ([5]) + RF model; a MiniRocket instance; a MultiRocket instance; arsenals for both MiniRocket and MultiRocket [6], [7]. Finally, we propose and develop a Hybrid Neural Network (HNN) architecture to try and surpass the scores of the pure ML models. This model combines a convolutional neural network (CNN's) and a transformer architecture to process the full sensor data from the conveyor systems.

Throughout the paper we will discuss and compare the efficacy of the aforementioned machine learning techniques and the HNN model for the multiclass anomaly detection task, as well as discuss the advancements made in preemptive maintenance and system health analysis. Our overall results were promising, with the HNN model surpassing all accuracy scores in comparison to the other methods (about 95.5%), as well as being within reasonable computation time for classifying multiple observations (approximately 0.4 seconds per observation).

II. Problem Description

Misalignment and pin idlers are the anomalies that most impact conveyor belt availability times, both in terms of occurrence frequency as in terms of maintenance delays [3]. Considering the need for belt replacements and repair times, belt failures end up being responsible for significant operational losses and overall increases production costs. The resulting damage from different types of anomalies range from mild material spillages to dangerous' belt physical stresses, which can cause ruptures to happen and in some cases pose severe risks for personnel. It is also important to note that ruptures lead to immediate halts to production, resulting in financial losses due to maintenance delays [2].

These failures may occur due to various factors, such as heat dilation, strong winds, and decentralized feeding of heavy ores, among various others [8]. Another factor that may lead to belt misalignment are problems with the belt's idlers (also called rolling pins). Figure 1 compares a belt in a normal position and a misaligned belt due to a misaligned pin.

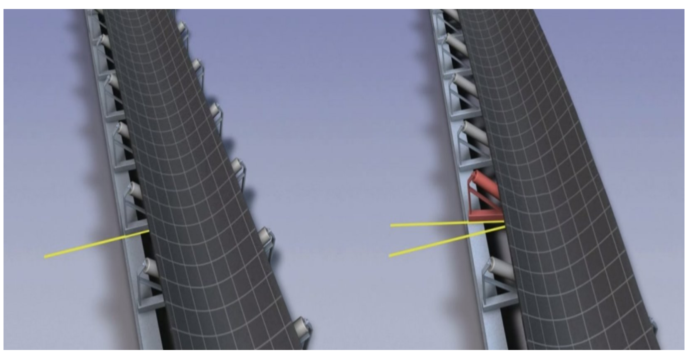


Fig. 1: Comparative of an aligned belt on the left and a misaligned belt on the right.

Current anomaly detection methods [3] for this task are primarily limited to single-observation detection, which fails to consider the sequential nature of the sensor data, leading to insufficient predictive capabilities. Additionally, multiclass anomaly detection, crucial for accurately identifying specific types of roller issues, is notably absent in existing approaches, specifically with regard to industrial health monitoring scenarios. The approaches that rely on time series lack sufficient accuracy, which highlights the need for a more robust solution capable of handling different anomaly types.

Regarding data availability, existing datasets, such as the TimeEval dataset [9] and the UCR time series repository for anomaly detection [10], often do not cater to the specific dimensionality needs of industrial conveyor systems. These datasets are typically one-dimensional, lacking the multi-faceted approach required for 3D space sensor data interpretation. Furthermore, the ones that do fit the format criterion are often restricted in scope, posing challenges in training models for specific and large-scale industrial applications. Another challenge lies in the variability of data quality and frequency. Sensor data often comes in different sampling frequencies, necessitating interpolation for standardization before analysis. This adds an additional layer of complexity to model training and accuracy.

The necessity for multiclass anomaly detection in conveyor belts cannot be overstated. Each type of anomaly requires a specific response; for example, a roller slightly out of place may have a higher level of concern than a roller that is not in contact with the belt. Accurately discriminating between these nuances is crucial for effective maintenance, as it directly impacts the decision-making process for repairs and replacements. Thus, a model capable of precise multiclass anomaly detection is essential for optimizing operational efficiency in industrial environments.

III. Related Works

Previously mentioned mechanical devices, such as switches, are commonly installed next to conveyor belts to detect when they deviate from their normal path. However, these devices can only partially solve the problem by alerting or turning off the equipment when a misalignment is detected. This may not be enough to monitor or prevent a conveyor belt fault completely. [2].

Some researchers have focused on monitoring conveyor belts as a part of their work. Farhat et al. [1] proposed an analysis of the motor current signature of the conveyor belt (MCSA) based on the fact that belt attrition, caused by mistracking, increases the motor load, resulting in current alterations. Ericeira et al. [11] evaluated the condition of idlers using ultrasonic sensor data. Different machine learning algorithms were applied to differentiate between problematic and non-problematic idlers.

Nienhaus et al. [2] proposed the use of thermographic cameras on Long Wave Infrared (LWIR) to monitor conveyor belts during feeding. The data acquired through this process was utilized to optimize the belt feeding system to prevent misalignment or to correct existing deviations in the belt's trajectory.

Chamorro and collaborators [12] proposed a method to monitor the condition of a conveyor belt. It uses a series of sensors to estimate the misalignment of the belt, which is achieved through machine vision techniques. The captured images are sent to a computer, where they are processed to evaluate the behavior of the belt. Žvirblis *et al.* [13] used strain gauges data, long short-term memory (LSTM), and Transformer neural network models were used for the classification of different conveyor belt conditions (loaded and unloaded).

Kirjanów-Błażej *et al.* [14] presented a method to evaluate the thickness measurement of conveyor belts. The method was based on using several ultrasonic sensors installed on conveyor belts. Cieplok [15] provided a study about dynamic damper applied for vibration analysis of the conveyor belts. The dynamic damper improves the vibration studies through anti resonance effect to achieve a significant decrease in vibration amplitude and forces transmitted to the foundation.

A monitor system based on UHF RFID sensor response is used for structural health monitoring by Zohra *et al.* [16]. However, the belt material affects the sensor performance because of the dielectric properties and rubber thickness. A study with a similar approach is proposed by Salim *et al.* [17]. It is about a monitory system based on UHF RFID sensor, which can provide a safer monitoring of the conveyor belt through data collected via RSSI (Received Signal Strength Indicator) from the sensors.

Zimroz *et al.* [18] suggested using IMUs to monitor conveyor belts. They did experiments in a reduced-scale conveyor belt with a coupled IMU. It was verified that idlers defects diagnostics and straightforward interpretation are signals of z acceleration and gyroscope in the Y direction. IMUs were also used by Yasutomi and Enoki [19]. A method to estimate the position of the IMU on a conveyor belt is proposed. A Deep Convolutional LSTM neural network was implemented to detect features that may inform position. Despite using IMUs to monitor conveyor belts, the aforementioned works do not measure inertial data during anomaly conditions.

In our previous work [3], we have used IMUs installed at the lateral extremities of the belt to evaluate its health along its route. Various tests were conducted on the conveyor belt, including its correct performance, structural anomalies, and misalignment under different work scenarios. However, we only used similarity analysis techniques to evaluate the conveyor belt's conditions. The comparison of the related works can be seen in Table I.

In our current work, we propose a more advanced analysis, using the previous experiment's methodology, that utilizes machine learning techniques such as a hybrid transformer architecture and random forests models alongside other temporal series feature extractors like Catch22, MiniRocket, and TSForest.

IV. Background and Proposed Architecture

As previously mentioned, we used a selection of machine learning and feature extraction techniques to classify the conveyor belt status. We initially experimented and trained a series of random forests, as well as models based on feature extraction, namely: the catch22 feature set, MiniRocket and MultiRocket arsenals, as well as a Time Series Forest (TSForest) model. We concluded our research with the development of a HNN network, fusing a CNN and a Transformer architecture that surpassed the accuracy of all previously tested instances. In the following subsections, we will present an overall view and discuss the tested models.

A. Random Forest

It is widely employed for solving large data regression and classification problems. A Random Forest algorithm consists of an ensemble of decision trees, each generated randomly [20]. We used 100 trees for the random forest models and the time series sub sequences as input without feature extraction.

B. Feature extraction-based models

Feature extraction-based classification, instead of using raw numerical data points as input, uses extracted statistical and domain characteristics from the raw data and uses these processed values as features. Examples of possible characteristics to extract are the mean, standard deviation, skewness, slope, and peak positions. However, features can also be extracted through more sophisticated methods. The following feature extractors algorithms were implemented and tested on our data:

1) Catch22 : The Canonical Time-series Characteristics is a set of 22 features designed for the analysis of time-series data [5]. These features capture various aspects of temporal patterns, including regularity, complexity, and periodicity. The term "Canonical" emphasizes their standardized applicability across different domains. Researchers and practitioners leverage Catch22 for diverse applications, such as signal processing, financial analysis, and healthcare. The reference [5] serves as a source for more detailed information on the Catch22 feature set.

2) Rocket algorithm family: In our study, we adopted an ensemble approach, utilizing both MiniRocket [6] and MultiRocket [7]. These algorithms represent modified versions of the Rocket algorithm, enhancing their efficiency and effectiveness in specific feature extraction scenarios for time series data. Both are convolution-based feature creation models, coupled with subsequent application of linear classifiers, forming a robust methodology for predictive modeling and classification tasks. The ensemble strategy incorporating MiniRockets and MultiRockets contribute to improved model performance by leveraging the strengths of both algorithms. For further details on the optimization and implementation of these algorithms, refer to the original papers [6], [7].

3) Time Series Forest (TSF): The TSF algorithm is an adaptation version of the Random Forest classifier [20] designed to handle sequential temporal data [21]. It divides time series into segments and employs decision trees to build models within each segment. The final class of the time series is determined by aggregating individual tree predictions through a voting mechanism. This approach enables TSF to effectively capture intricate temporal patterns, making it a versatile tool applicable to various temporal data scenarios. For more details on TSF's methodology and applications, refer to the foundational works [20], [21].

C. Hybrid Neural Network architecture

Finally, we propose the use of a hybrid architecture, which fuses a 1D convolutional neural network (CNN) as the input network and a transformer architecture as the output network to process the time series data to correctly classify and differentiate anomalies.

TABLE I: Comparison of the related works.

Study	Current Signature Analysis	Ultrasonic Sensor	Camera	RFID	Strain Gauge	IMU	Anomaly detection
Farhat *et al.* [1]	✓	✗	✗	✗	✗	✗	✓
Ericeira *et al.* [11]	✗	✓	✗	✗	✗	✗	✓
Nienhaus *et al.* [2]	✗	✗	✓	✗	✗	✗	✓
Chamorro *et al.* [12]	✗	✗	✓	✗	✗	✗	✓
Žvirblis *et al.* [13]	✗	✗	✗	✗	✓	✗	✓
Kirjanów-Błażej *et al.* [14]	✗	✓	✗	✗	✗	✗	✓
Salim *et al.* [17]	✗	✗	✗	✓	✗	✗	✓
Zohra *et al.* [16]	✗	✗	✗	✓	✗	✗	✓
Yasutomi and Enoki [19]	✗	✗	✗	✗	✗	✓	✗
Zimroz *et al.* [18]	✗	✗	✗	✗	✗	✓	✗
Matos *et al.* [3]	✗	✗	✗	✗	✗	✓	✓

1) 1D Convolutional Layers:: Comprised of three sequential layers, the 1D CNN network employs 256 kernels in each layer. The use of such architecture as the input network involves reducing multidimensional time series data into smaller matrices while maintaining contextual and positional information for each observation [22]. Through this process, the model extracts temporal features and captures contextual relations among the multidimensional features.

2) Transformer Architecture:: Following the convolutional layer, we employed a transformer architecture [23] with 32 attention heads and 4 layers. The transformer architecture takes as input the intermediate features processed by the CNN, so it receives the condensed time series data and uses the attention mechanisms to classify the processed sequences.

The hybrid nature of the model offers several advantages. Firstly, it provides a level of interpretability [24], with the convolutional layers offering context-aware data processing and the transformer providing insights into sequence relationships. The training required, on average, 2 to iterate 500 epochs with a batch size of 64 series. We used a learning rate of 1e-3, and the Adam Optimizer [25] to train the model.

V. Methodology

The laboratory experiments were conducted using a reduced-scale conveyor belt and an NGIMU sensor from X-io Technologies. The NGIMU sensor was positioned at the extremities of the belt to monitor its behavior. The sensor axes work as following:, the belt is a flat surface, with the X axis starting at the final end, traversing along the length of the belt, the Y axis perpendicular to the belt, and the Z axis orthogonal to the plane of the conveyor belt (pointing to the center of the earth). The NGIMU, an inertial measurement unit, transmitted data via Wi-Fi network and with the User Datagram Protocol (UDP). The conveyor belt used in the experiment had five pairs of idlers, each measuring 110 cm in length and 25 cm in width.

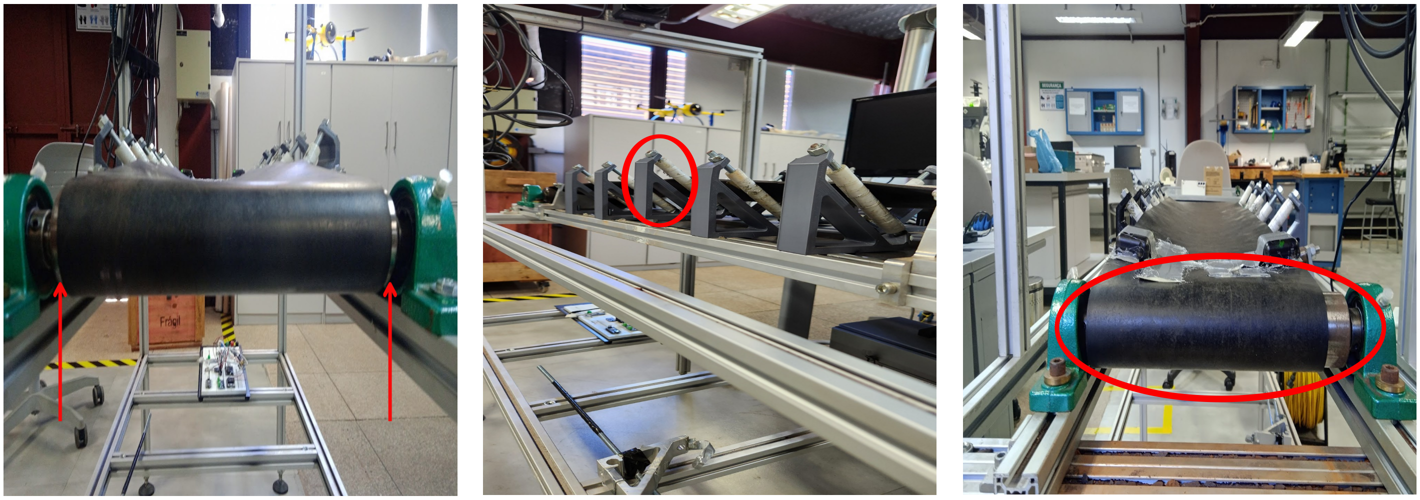
We have simulated different work scenarios for the conveyor belt. Initially, we gathered accelerometer and gyroscope data while the belt operated under normal conditions (i.e., aligned), as depicted in Figure 2a. These data serve as Ground Truth or references for the other scenarios' data. Then, we created abnormal scenarios, specifically on the belt's idlers. For the idler anomaly, we raised the third idler's position, first on the left and then on the right. Figure 2b displays the elevated idler on the left. The other anomaly scenarios generated were more drastic: we removed the left idler first, and then the right idler to simulate a severe anomaly.

We collected IMU data for each scenario, with the conveyor belt speed set at 2.25 m/s and the acquisition frequency varying from 100 Hz to 500 Hz for the accelerometer and gyroscope. The resulting dataset had a total of 156 experiments, with different time series sizes. Each experiment was interpolated to a size of 500 observations, with six variables (three for the 3D accelerometer and three for the 3D gyroscope).

Analyzing the raw measurement data can be difficult due to the noisy and asynchronous nature of the IMU signals. To address this issue, we have developed an algorithm that utilizes previously collected raw measured data. This algorithm is designed to detect data within the range of interest (ROI) by identifying signature data that occurs when the sensor moves from the superior to the inferior side of the belt. By doing so, it retrieves only the measured data taken when the NGIMU was on the superior side, when the sensor is in contact with the belt's idlers. This approach helps to ensure that the data collected is accurate and reliable.

Before classifying the time series, we split up the series we received from collecting the signals on the model. To do this, we used the peaks generated by the gyroscope's Y axis to separate the signals collected only on the upper part of the belt, because the anomalies generated were all on the rollers that support the belt, so the data from the lower part was removed along with the signals from the end of the belt, when the sensor reverses position. We also applied a moving average to smooth out the series and make the task of separating the signals easier. After dividing the time series, we interpolated it with a linear spline so that all the series to be classified have the same number of points.

To validate each model, we used Leave-One-Out Cross-Validation (LOOCV), a cross-validation technique in which each data point in the set is used as a test set exactly once. The model was trained with the other points for each data point and evaluated on the point left out. Performance was evaluated



(a) Align Conveyor belt. (b) Elevated idler on the left. (c) Belt deviation to the left.

Fig. 2: Experimental scenarios.

for each iteration, and the results were averaged to estimate the model's performance. Although it can be computationally costly, LOOCV is useful on small data sets, maximizing the use of data for training. This technique was chosen because our dataset was small and the computational cost was not excessive. We have analyzed the right sensor data. However, the validation process is analogous to the left sensor.

VI. Results

Table II compares each developed model. The Hybrid Transformer model showcased commendable performance in anomaly detection on our test conveyor belt system. The Key performance metrics are accuracy, precision, recall, and F1-score all around 95.55%. These results demonstrated the model's overall effectiveness. Moreover, a AUROC score of 99.76% highlights the model's superior capability in distinguishing between normal and anomalous states.

Figure 3 shows the confusion matrix of the hybrid transformers model. The model was capable of classifying most of the features. However, it experienced difficulties in detecting idler's faults on the opposite side. Figure 4 shows the ROC curve obtained on the development of the HNN.

Models based on feature extraction performed well in terms of accuracy, especially ensemble strategies such as MultiRocket and MiniRocket Arsenal. Of particular note is the MiniRocket, which sim than the HNN and MultiRocket. The

TABLE II: Comparison of Model Performance

Model	Accuracy (%)	AUROC (%)
HNN	**95.55**	**99.76**
Multirocket	91.02	94.27
Mini Rocket	89.10	93.05
TSforest	85.25	90.57
RF	83.97	90.50
Catch22 + RF	76.28	85.38

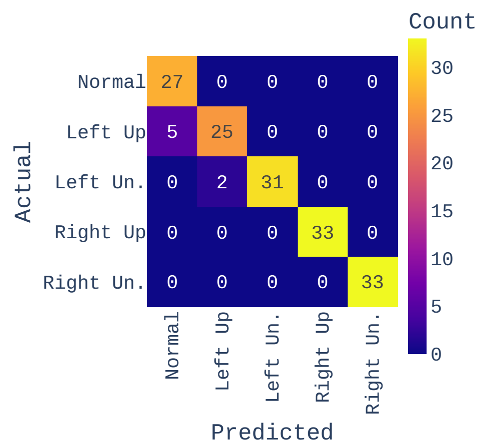


Fig. 3: Confusion Matrix for the HNN model.

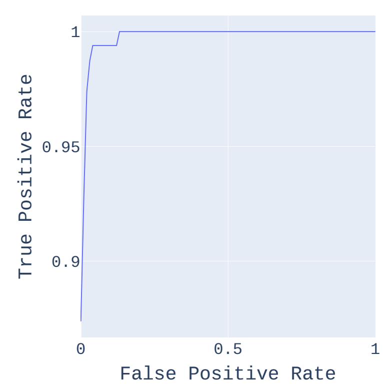


Fig. 4: ROC curve averaged over all classes for the HNN model.

others provided inferior results, but they are simpler models that don't use the ensemble strategy or even extract features like RF. One observation is that catch22, a feature extractor, performed worse than RF without feature extraction. However, the Time Series Forest algorithm, which extracts features from sub sequences, proved to be a more interesting way of classifying our time series, with better results than RF, classifying the series with features extracted from the whole series with catch22 and also RF classifying the series with the raw data. In summary, the Hybrid Transformer model's performance demonstrates its potential as a robust tool for predictive maintenance in industrial conveyor belt systems, demonstrating a marked advancement over existing anomaly detection methods.

VII. Conclusions

This paper proposed a method for evaluating the health of conveyor belts using two inertial sensors and machine learning applied to a time series. We tested diverse approaches to classify the conveyor belt behavior. Machine learning techniques such as Random Forest, hybrid transformers, and models based on feature extraction were experimented.

In an experiment, we installed IMUs in a reduced-scale conveyor belt and collected reference and anomalous data by altering the device structure. We used machine learning multiclass classification models to classify the conveyor belt health. On the comparison of each algorithm, the hybrid transformers model obtained a 95.55% accuracy, higher than the other experimented models.

Using the developed model, we could detect the conveyor belt's idler anomalies on both sides, even utilizing only one side IMU data therefore solving our initial problem of improving on existing multiclass anomaly detection algorithms. As part of our future research, we aim to collect a bigger database on an industrial conveyor belt and implement the machine learning strategies mentioned in this study.

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